

D-LIVER

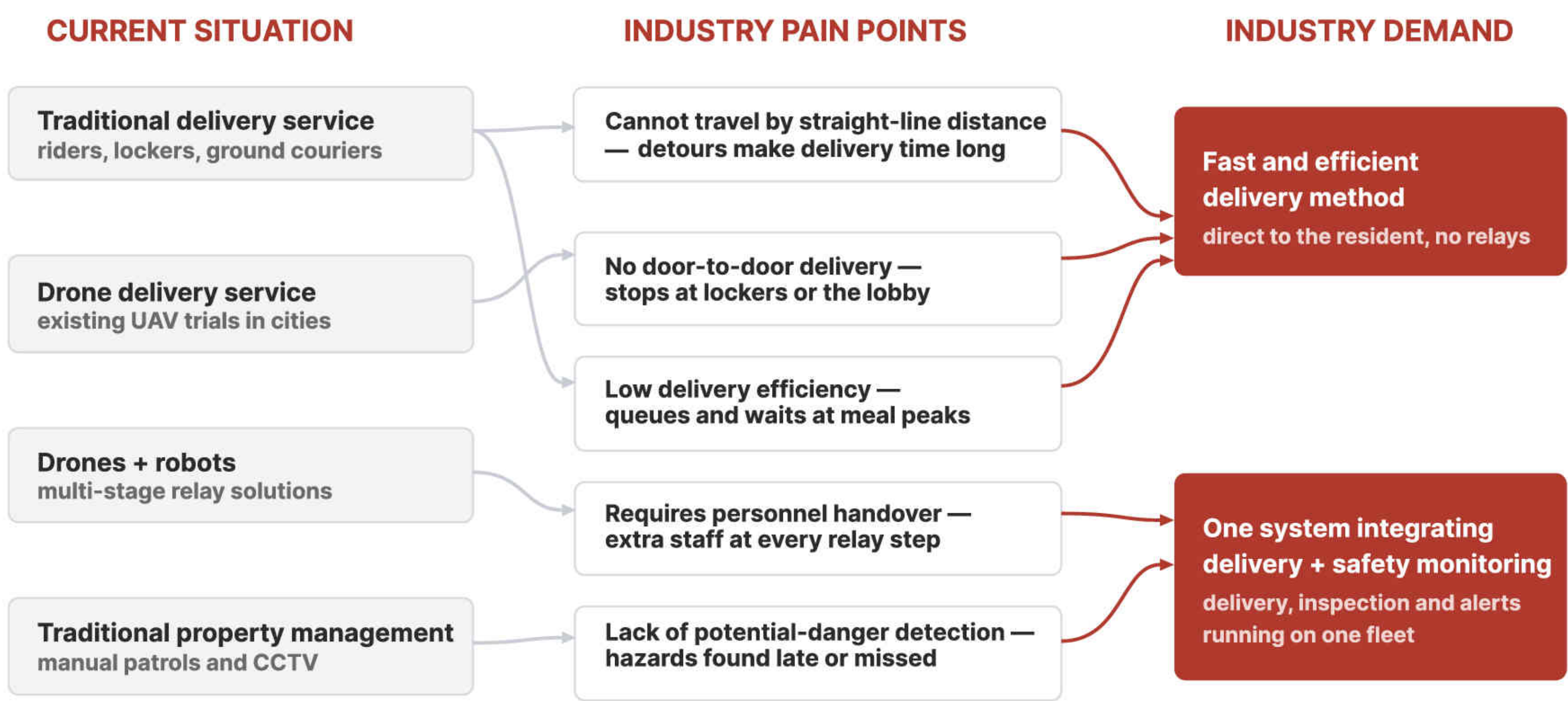
Drone-Assisted Community Living

Window-to-window delivery, aerial inspection and emergency response for high-density Greater Bay Area communities.

LATX7512 · Group 1 · Product Proposal



THE PROBLEM



Every current option breaks somewhere on the last 100 metres — and none of them watches the community while it works.

Current options vs. pain points vs. demand — every path breaks on the last 100 m

ROLLOUT ① Pilot in 1-2 tech-friendly estates under the regulatory sandbox → ② merchant & property-manager expansion → ③ GBA network + government co-operation.
BUSINESS MODEL B2B2C — property managers are customers, residents are users. Revenue: delivery fees + system access + hardware rental

Inefficient "last 100 m" — riders cannot enter buildings; residents walk down or wait for re-delivery; lockers overflow at meal peaks. Manual patrols — delayed issue detection, unsynced information, slow maintenance with unclear ownership. Policy tailwind — HK's 2025 low-altitude-economy sandbox + dedicated 1.4 GHz spectrum make drones a regulatory priority; "smart community" already sells premium estates.

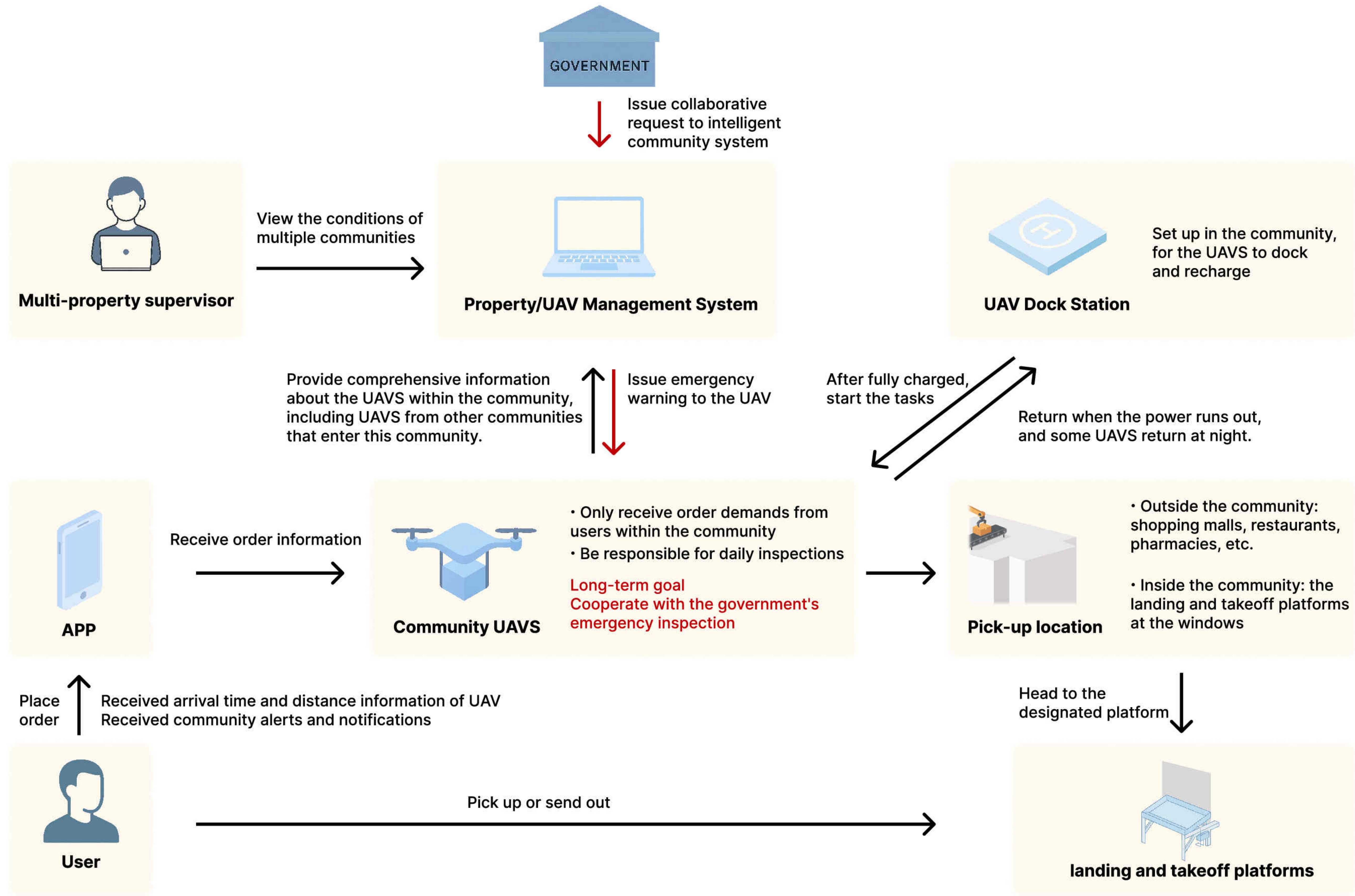
MARKET — GREATER BAY AREA STRATEGY

One playbook, two anchors: validate value in HK, iterate cheaply in SZ, then replicate across bay-area island and waterfront communities.

Hong Kong · value Highest order value — ARPU ≈ US\$1,001 (2024). Premium estates pay for smart services; outlying-island estates depend on ferry logistics — underserved early demand.	Shenzhen · speed Most pilot-friendly regulation and a mature drone industry chain; urban UAV meal delivery already routine — the fastest place to iterate and cut cost.
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HK food delivery: **6.86% CAGR → US\$6.15B by 2030 · 59.6% user penetration by 2026 · 38 sandbox pilots · 1.4 GHz drone spectrum**
Sources: Statista · HK LAE regulatory sandbox 2025 · diagrams, UI & renders from the D-LIVER group deck

SYSTEM ARCHITECTURE — HOW EVERY ROLE CONNECTS



Government requests & emergency warnings flow through the Property/UAV Management System; UAVs serve orders only within the community, handle daily inspection, and recharge at dock stations. Users order via App and collect at window landing platforms.

COMPETITIVE ANALYSIS

Three substitute paths to the same user problem

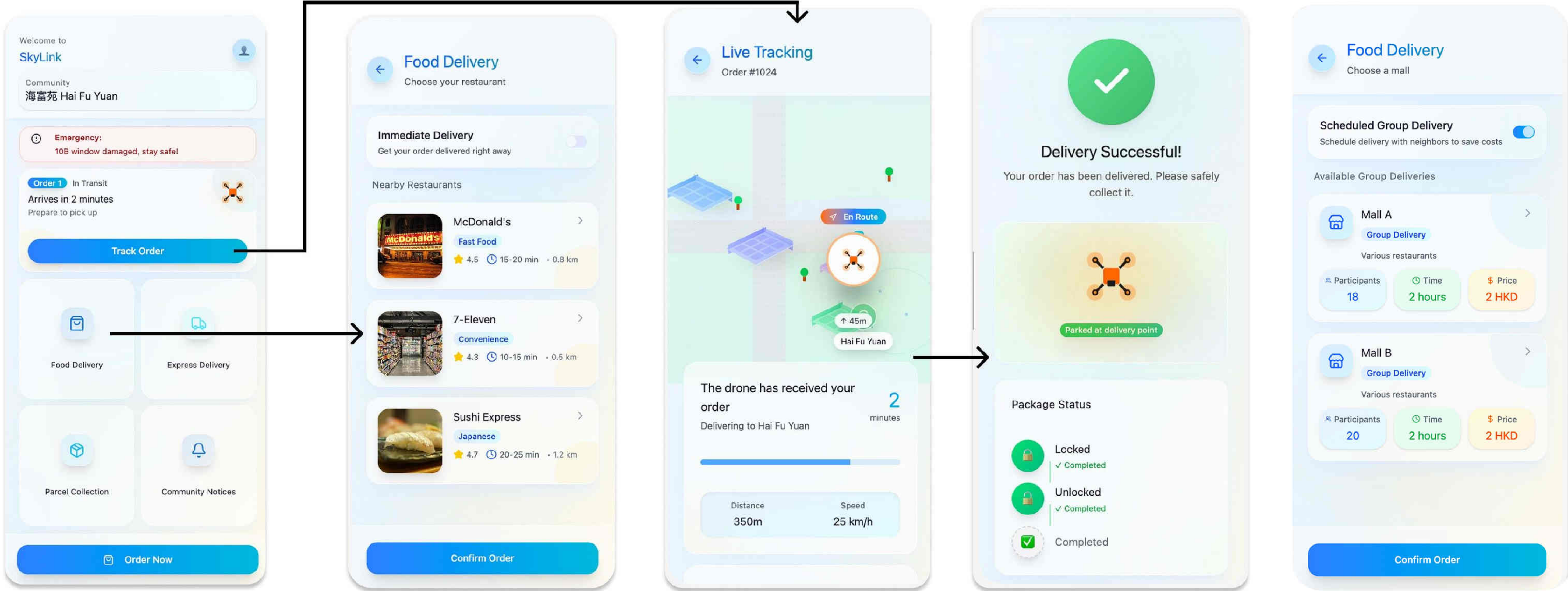
		
Meituan — platform + UAV trials	Wing — pro drone delivery	Longfor-type — smart PM
GAP		
Rider → station → locker chain; no in-building reach	Needs open landing zones — poor fit for HK and Shenzhen	No last-meter delivery, no aerial inspection

Edge: D-LIVER alone combines platform reach, drone logistics and property integration — window-direct, two jobs on one fleet, built for plot ratio > 10.

Product Design: App · Web Platform · Hardware

Four modules on one closed-loop platform — Delivery & Shipping · Intelligent Inspection · Emergency Alert · Equipment Management

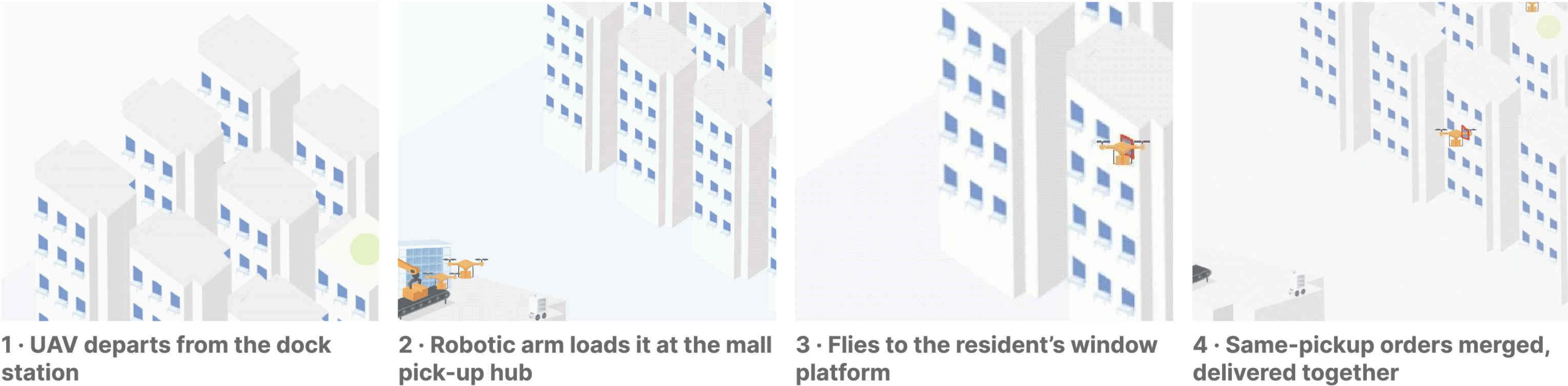
RESIDENT APP — ORDER TO WINDOW IN MINUTES



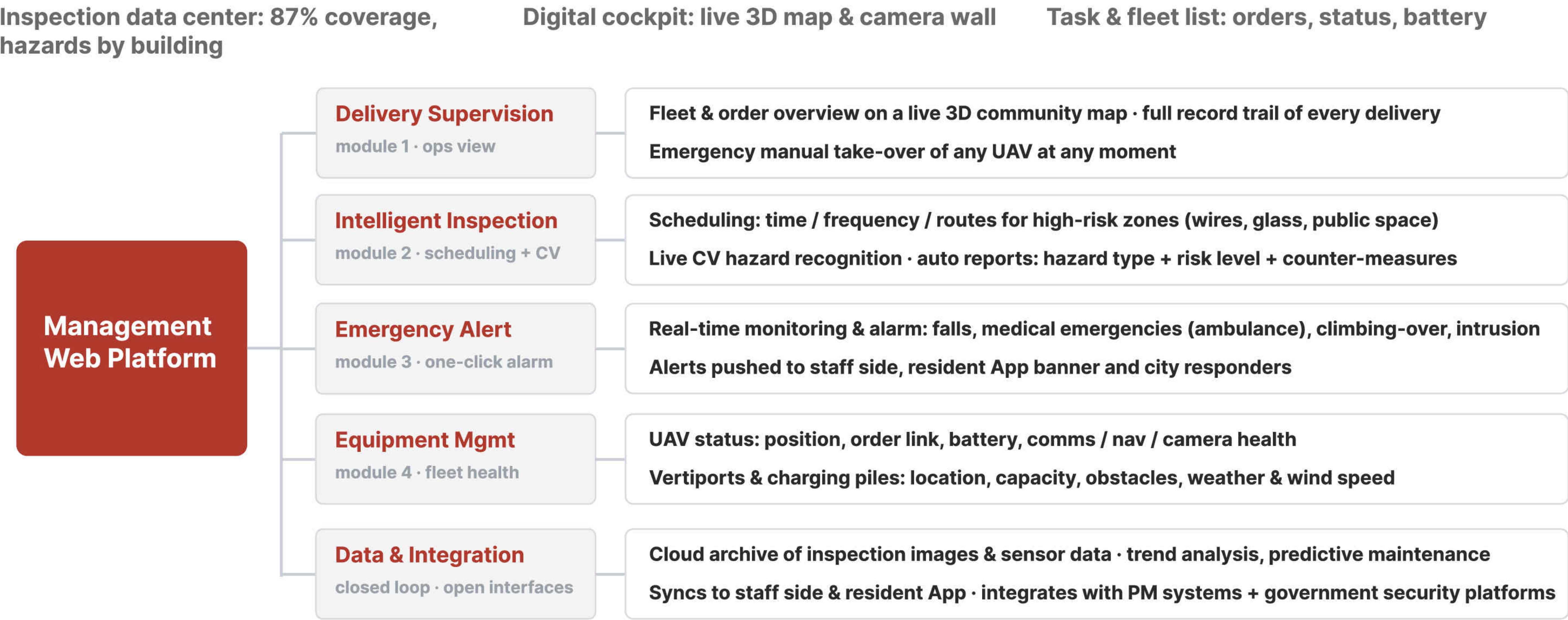
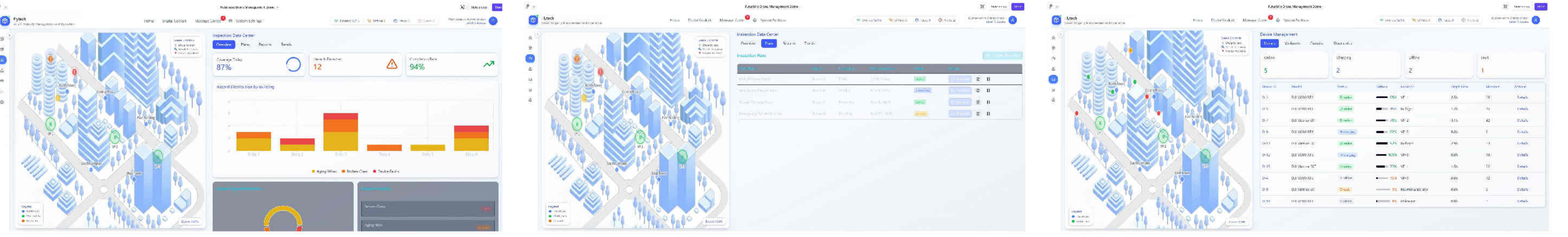
Module 1 · Delivery & Shipping — orders from the same pickup point are merged; one flight serves multiple windows across floors. In range → direct window drop; out of range → mother-child drone relay.

PERFORMANCE ≤ 25 min per sortie · path –15–20% · 100% patrol coverage at ≤ 30 min intervals · planning ≤ 3 s · correction ≤ 0.5 s · 3-tier path system (global – local – dynamic) · IoT temp-box 6–60 °C

DELIVERY PROCESS — FROM DOCK TO WINDOW



MANAGEMENT WEB PLATFORM



HARDWARE — TWO DESIGNS, LOGIC AS FLOW

